



A stochastic chance-constraint framework for poultry planning and egg inventory management

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Received: 8 August 2023 / Revised: 1 July 2024 / Accepted: 4 July 2024 / Published online: 29 July 2024
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Abstract

This study addresses the capacitated lot-sizing problem in the poultry industry for egg production planning, aiming to minimize production, transportation, and inventory costs. This problem has already been investigated with data certainty and formulated as a mathematical model and a heuristic algorithm has been applied to solve it due to high complexity. In this study, we reformulate the same problem as a new mixed integer linear programming model to achieve optimal solution in a relatively short time without the need for heuristic algorithms. To evaluate the model performance, it is executed using the available data, and its efficiency is validated by comparing the obtained results. Subsequently, the uncertainty of weekly demand is considered, leading to potential shortage or surplus in storage. To address this uncertainty, the chance-constraints method is employed with various attitudes, and several production plans are proposed accordingly. The performance of these plans is compared using random data, and the most suitable programs are identified. The presented decision-making tool can provide production planning that meets customer demand with high reliability while also minimizing surplus inventory in the warehouse.

Keywords Lot-sizing · Poultry industry · Egg production planning · Demand uncertainty · Chance-constraint

1 Introduction

Sustainable supply of food in any society is a basic condition for the survival of that society, and for this reason, a very detailed planning for production or import, as well as proper storage and distribution of food needed by the people must be undertaken. To guarantee the timely and quality supply of food, the supply chain must be effectively managed from raw material procurement to processing, storage and distribution of the final product. This is necessary to prevent any disruption in fulfilling the citizens' demand. The need for accurate planning in food supply chains can be seen from two perspectives. The first is the strategic importance of many food items, as the lack of these items can lead to public discontent and disrupt societal order. The second is the perishability of food. When considering the problem from the first perspective, it is important to provide and store adequate food supplies on time to meet demand. From the

second perspective, it is crucial to be conservative in production and storage to prevent food spoilage. Therefore, it is very important to provide detailed plans for the production, storage and distribution of food that take these two points of view into account in their design.

Poultry industry is one of the important branches of food industry whose products including meat and eggs are in high demand. A poultry farm is a place that raises chickens specifically for either meat (as broilers) or egg production (as layers). The farm has separate facilities where the birds are sheltered, fed, and nurtured. To lay eggs, they need a healthy and balanced diet that includes protein, vitamins, and minerals. Maintaining uniformity and good quality egg production in these farms throughout the year can be challenging. This involves proper nutrition, sufficient lighting, appropriate temperature, and regular sanitation. Disease outbreaks, insufficient ventilation, and stress can also hinder egg production. Therefore, farmers must have a thorough understanding of poultry management and implement best practices to ensure consistent production of high-quality eggs. The following steps should be properly managed in any laying hen farm. Spawning until the birth of chickens, rearing immature chickens in suitable halls until they reach

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maturity, keeping and feeding mature laying hens in larger halls and finally transferring old chickens to the slaughterhouse. There is a time gap between each of these stages, and the lack of proper coordination of these steps can lead to a continuous interruption in the production and supply of eggs or overproduction and excess spoilage. Therefore, it is necessary to utilize a suitable scientific tool that can accurately determine the amount of egg production in different periods. The output planning should enable us to meet the demand of each period with confidence while ensuring that the produced eggs are sold before they spoil. The amount of hatching is dependent on the amount of demand for eggs in the following weeks (because it takes several weeks from hatching to egg laying). Therefore, we need a suitable planning that can determine the amount of egg production in different periods in such a way that we can meet the demand of each period with high confidence and at the same time be sure that the produced eggs will be sold before they spoil.

This problem can be defined as the multi-period lot-sizing problem. The multi-period lot-sizing problem, in its simplest form, is defined as follows: the demand for a product is given over a multi-period horizon, and the amount of production or product order in each period should be determined in such a way that it meets the total demand and minimizes the sum of fixed and variable costs of order/production, as well as other costs such as inventory holding (Wagner and Whitin (1958)). This problem has numerous aspects and assumptions, and researchers have explored various types of these problems based on the number of items and warehouses, the type of objective function, and various solution methods for developing applications of this problem (Wagner and Whitin (1958), Aouadni et al. (2019), Buschkühl et al. (2010), Jans and Degraeve (2007)).

Generally, in such problems, finding an optimal solution becomes increasingly difficult as the problem size increases, and obtaining an optimal solution with larger data sets becomes a challenging and practically impossible task. For this reason, heuristic and meta-heuristic algorithms have been developed to obtain a suitable answer with less CPU time (Jans and Degraeve (2007)). The multi-period lot-sizing problem includes various problems for example, see (Guner Goren et al. (2010), Witt (2019), Chen (2015), Alfares and Turnadi (2018), Gansterer et al. (2021), De Armas and Laguna (2020) and Mediouni et al. (2022)).

One of the most essential and challenging issues in the food industry is proper planning for efficient and cost-effective management of activities to meet demands on time while reducing production costs and preventing spoilage of surplus foods. To address this significant issue, researchers have proposed various planning processes to minimize production costs and improve the management of food industry activities. For example, (Abedi and Zhu 2017) presented an optimization model for purchasing spawns, the production

process of fish breeding, and the distribution of harvested fish in the fish supply chain. They presented a production planning optimization model using mixed integer linear programming (MILP) to maximize total profit. They considered the main characteristics of fish farming, including loss rate, value-added chain, fish weight, and fish age. Mediouni et al. (2022) developed a MILP model for the lot-sizing and scheduling problem in the dairy production process. They solved this model using exact methods and a relax-and-fix heuristic method for some data samples. Sethanan et al. (2013b) proposed a model for planning chicken orders and production scheduling in a poultry farm company. This involved allocating pullets to farms, assigning egg-laying chickens to farms, and transferring them to slaughterhouses. They used genetic algorithms and local search methods to determine the number of pullets and poultry farms needed to solve the problem. In another study, Sethanan et al. (2013a) developed a heuristic algorithm to solve the problem of scheduling egg production in chicken farms. They used Wagner and Whitin's heuristic algorithms to determine the appropriate number of chickens to replace birds with reduced egg-laying capacity. Boonmee et al. (2015) proposed a hybrid growing neural gas approach algorithm to minimize the total cost of allocating chickens to chicken houses. The total cost includes operating the farm, transporting chickens, and the loss from mixing hens at different ages in identical hen houses. Boonmee and Sethanan (2016) proposed a multi-level capacitated lot sizing problem for egg production in the poultry industry. This model is a mixed integer computing and programming tool for solving small-size problems. They further developed particle swarm optimization to solve problems with large data sets. In this issue, the goal is to determine how many chickens to lay and which of the chicken farms to meet customer demand with the least cost in the case of the demand parameter being certain. Because the age of the chickens is included in the modeling of this article, therefore, this issue has high computational complexity. Brevik et al. (2020) proposed a MILP model for integrating hatcheries and farms in the broiler industry. They used a meta-heuristic rotating horizon algorithm to solve it. Solano-Blanco et al. (2023) proposed a MILP model to optimize broiler production and inventory decisions. The model aims to meet wholesale-level demand and operating constraints while providing optimal decisions for the farm and slaughterhouse.

Data uncertainty is an inherent component of many real-world problems, and ignoring uncertainty makes proposed programs impractical or inefficient. In production planning, the level of demand plays a crucial role in determining the timing and volume of production. In certain industries, there may be a significant gap between the decision-making process to determine production levels and the actual determination of demand, making it challenging to accurately predict demand. For example, in order to meet the demand

for eggs, we have to raise the right number of chickens a few weeks earlier so that they lay enough eggs at the right time. Estimating the amount of egg demand during hatching is not an easy task. If production levels fall short of demand, it can lead to inventory shortages and harm the company's reputation. On the other hand, overproducing can result in excess inventory and potential financial losses, both of which can impose significant costs on the company. Therefore, in production planning, it is important to treat the demand parameter as a random variable and anticipate different scenarios. To deal with the data uncertainty conditions in mathematical models, various viewpoints and methods have been presented, among which stochastic programming, fuzzy approaches, chance constraints methods, and robust optimization can be mentioned. Researchers often use some methods such as fuzzy optimization in which the uncertain parameter is placed in a set with a certain membership function. For example, Ali et al. (2020) proposed a fuzzy multi-objective model on a multi-stage reverse logistics network for product retrieval and Ali et al. (2021) investigated the complexity of multi-product, multi-location supply chain issues. They considered maintenance costs, purchase costs, and demand parameters as uncertainty and used fuzzy optimization to deal with them. Some other researchers have used robust optimization to deal with uncertainty where the uncertain parameter is placed in a certain set. Dadaneh et al. (2023) considered the demand of customers in an interval and used robust optimization to manage this uncertainty. Goli et al. (2023) considered a flexible, sustainable, multi-product, multi-period supply chain network as an integrated forward/reverse logistics system. They considered the demand parameter uncertainty and used robust optimization to deal with it. Another approach to dealing with uncertainty is stochastic optimization of service-level constraints. Pauls-Worm et al. (2016) studied the decision-making problem of producing fresh food products with long lead times, considering the age of the production items and the uncertain demand for these products. They modeled the problem as stochastic programming and used a β -service level approach to address demand uncertainty. Tempelmeier and Hilger (2015) addressed the capacity-constrained stochastic dynamic lot-sizing problem for multiple items under two types of fill rate constraints, solving them with a variant of the fix-and-optimize heuristic algorithm. Tavaghof-Gigloo and Minner (2021) studied the stochastic lot-sizing problem and introduced a MILP model with service-level constraints for producing final products with uncertain demand directly from raw materials using a capacity-constrained machine.

If the distribution function of the random variable is known, the chance constraints method is a suitable tool to handle it. Chance constraints programming was introduced by Charnes et al. (1958) and Charnes and Cooper (1963) and has become an important and effective method for

dealing with uncertainty in supply chain issues involving data uncertainty. This approach has recently attracted the attention of researchers working on various issues, including (Karimi et al. (2021), Izadikhah et al. (2019), Abad and Pasandideh (2021), Zahiri and Pishvae (2017) and Zhang et al. (2023)). Pauls-Worm et al. (2014) addressed the scheduling problem for a food manufacturer facing non-constant demand for a perishable product with a fixed shelf life. They considered demand uncertainty and formulated the one-item, one-level production planning problem as a stochastic model with a chance constraint. The un-capacitated single-item dynamic lot-sizing problem with stochastic demands and backorders was examined by Tempelmeier (2007), who presented a model that minimizes start-up and maintenance costs subject to a chance limitation and a fill rate restriction. Gicquel and Cheng (2018) addressed the problem of determining the capacity lot sizing of a single-item resource with stochastic demand, modeling it as a chance-constrained problem where the probability of inventory shortages during the planning horizon is limited to the maximum acceptable risk level. Esra (2020) studied the stochastic lot-sizing problem with probabilistic carbon emission constraints and developed a MILP formulation that solves the problem in reasonable times using the chance constraints approach. Zhang et al. (2019) considered a two-stage stochastic lot-sizing problem with a chance constraint and developed a branch-and-bound algorithm to solve it. These studies demonstrate the effectiveness of using chance constraints to deal with uncertainty in supply chain problems, making them a valuable tool for decision-making in this field.

The uncertainty of the demand parameter in food inventory management is greater than in many other industries. This is because, on one hand, ensuring a steady supply of food to meet citizens' demands is a top priority, necessitating high production volumes. On the other hand, it is crucial to guarantee that the produced volume is sold before its expiration date. In the poultry industry, the amount of hatching depends on the demand for eggs, and the exact amount will be determined after a few weeks. Therefore, in modeling the problem, this parameter should be considered as a random variable. This approach can help reduce the potential problems resulting from its lack of estimation, such as deficiencies or excesses, as much as possible. As a result, many studies have proposed methods to minimize the costs resulting from inaccurate random data and provide programs with acceptable long-term performance or conservative programs that remain feasible for all predicted values of the stochastic data or with a probability of violating them less than a specific limit (Peidro et al. (2009)). Yazdekhasti et al. (2021) investigate the effects of the Covid-19 pandemic on the poultry industry by developing of a multi-period stochastic poultry supply chain. They have proposed two models aimed at dealing with the adverse effects of the pandemic through

product storage policy. In both models, poultry production is harmful and affected by Covid-19. Considering the complexity of the model, a hybrid solution approach based on branch-and-cut and dynamic programming is developed.

The research gap identified in this study can be divided into two parts, each of which has its proposed solution. In the first part, we aim to introduce an efficient mathematical model that can encompass all the assumptions put forth in previous similar study (Boonmee and Sethanan (2016)) while also maintaining lower computational complexity. In the second part, neglecting the uncertainty of demand at the time of hatching significantly impacts the output program derived from the presented deterministic model. It is crucial to consider the uncertainty of demand at the time of hatching (which occurs a few weeks before the laying age) to address this issue. The distinction between this concern and the scale issues encountered in other industries lies in the fact that each chicken raised at the laying age produces eggs continuously for at least three weeks, impacting the egg stock throughout this period. Therefore, an effective method is needed for scheduling the timing of hatching and producing the appropriate quantity of eggs to ensure that demand is met without shortages and to prevent surplus production, thereby avoiding waste confidently.

According to the research gaps stated above, this article pursues the following goals: Firstly, it presents a new mathematical model with low computational complexity that obtains the optimal solution for a real world case in a relatively short time. Secondly, it considers the uncertainty of the egg demand parameter as a random variable with a certain distribution. To address this uncertainty, stochastic programming and chance constraints framework are applied to determine effective planning. Here, effective planning refers to determining the number of hatchings in different periods in a way that prevents shortages or surpluses of eggs with high confidence. For this purpose, five production planning strategies are proposed and compared based on the fixed costs of the problem and the costs associated with shortages and surpluses of eggs. Consequently, the decision-maker can select the best plan based on the goals of egg production, customer demand, and associated costs. This research presents a new model for egg production planning in the poultry industry, formulated as a multi-level capacitated and multi-period lot-sizing problem to minimize total costs, which obtains the optimal solution in a relatively short time compared to the previous study. The weekly demand for eggs is considered as an uncertainty, and to deal with this uncertainty, five plans for egg production are presented with a new approach based on chance constraints. The article aims to achieve the following objectives:

- Providing an efficient model for the poultry industry and egg planning that considers all the conditions of the prob-

lem studied by Boonmee and Sethanan (2016) to reach an optimal solution in a short time. The presented model has low computational complexity and does not need to provide heuristic and meta-heuristic algorithms to obtain the optimal solution with the data of the previous article.

- Considering customer demand as random data and provide different methods to deal with uncertainty based on stochastic optimization and chance constraints.
- Implementing stochastic optimization to address customer demand uncertainty and presenting five stochastic programming methods with a new approach to deal with this uncertainty.

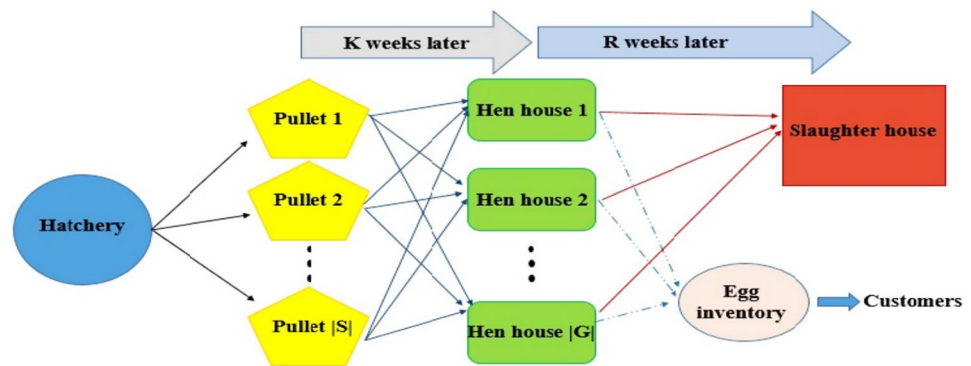
The article is structured as follows. Section 2 presents the problem definition and modeling under data certainty conditions. Section 3 presents a framework based on chance constraints for dealing with demand uncertainty, and five production plans from different perspectives are suggested. Section 4 presents the results of implementing the deterministic model and analyzing the stochastic methods outputs. Finally, Section 5 concludes the article.

2 Problem definition and deterministic modeling

The problem under consideration involves a poultry farm with a specific configuration, including pullet houses, hen houses, and a slaughterhouse with designated capacities. The distances between the pullet houses and each hen house are also determined. The process begins with ordered hatched chickens being delivered to some pullet houses and raised until reaching a specific age (K weeks). When they are mature enough, they are transferred to the hen houses, where they lay eggs at a specific rate (L_r) for a certain period (R weeks). After observing a decrease in egg-laying, the hens are transferred to the slaughterhouse ($R+K$ weeks after their arrival). Each week, the eggs produced in all hen houses are sent to the warehouse to meet customer demand. This process repeats periodically within a specified time horizon. The chicken and pullet houses are then out of service for a designated cleaning period. The number of ordered chicks per week depends on egg demand, the number and capacity of pullet and hen houses, and the capacity of the slaughterhouse. Figure 1 provides an overview of the poultry farm and egg production processes.

Planning on this farm involves deciding the number of chickens to be ordered each week, assigning these chickens to suitable pullet houses, and allocating appropriate hen houses for the adult chickens after K weeks so that capacity limitations are respected and costs related to ordering, transportation, storage, and inventory shortages are minimized.

Fig. 1 An overview of a poultry farm and egg production issue



Each process cycle, from hatching to transferring the chickens to the slaughterhouse, takes $R+K$ weeks. In this process, chickens start laying eggs from week $K+1$ and continue until week $R+K$, after which the chickens are transferred to the slaughterhouse for slaughter. After the young chickens (pullets) are transferred from the pullet house to the hen house, the chicken house becomes unusable for cleaning for a certain period. It is clear that the time from the arrival of the chickens to the chicken house until sending to the slaughterhouse is specific and fixed, and there is no need to influence the growth age and laying age in the decision variables, as in the model presented in Boonmee and Sethanan (2016). In the newly presented model, as compared to the previous similar model, the variables and restrictions that were used in determining the age of chickens and hens over the time horizon have been removed without disrupting the overall problem. This has led to a simpler way of modeling the problem of egg production and planning, with the newly presented model having low computational complexity and being able to obtain the optimal solution in a relatively short amount of time. A new formulation of the problem with deterministic demands is presented as follows. First, the symbols are introduced as follows:

Sets

S	Pullet houses ($s \in \{1, 2, \dots, S \}$)
G	Hen houses ($g \in \{1, 2, \dots, G \}$)
T	Time periods (weeks) ($t \in \{1, 2, \dots, T \}$)
A	Age of hens in the laying period (weeks) ($r \in \{1, 2, \dots, A =R\}$)

Parameters

K	The duration of keeping the chickens in pullet houses (weeks)
R	The duration of keeping the hens in hen houses (weeks)
Cfq	Fixed cost of ordering chickens
A^p	Fixed cost of operating a pullet house
A^H	Fixed cost of operating a hen house
K^p	Feed cost per chicken during K weeks

K^H	Feed cost per hen during R weeks
El	The weekly inventory cost per egg
KU	The cost of not meeting demand per egg
cd_{sg}	The cost of transporting a chicken from the pullet house s to the hen house g and then to the slaughterhouse
INV_{t_0}	Initial egg inventory at the beginning of the time horizon
DEM_t	Demand for eggs in period t
C_s^p	The capacity of pullet houses
C_g^H	The capacity of hen houses
YC	The weekly capacity of the slaughterhouse
L_r	The production rate of hens at the age r weeks
U^p	Cleaning time of a pullet house after the transfer of the pullets to the hen houses (weeks)
U^H	Cleaning time of a hen house after the transfer of the hens to the slaughterhouse (weeks)

Integer variables

H_{st}	The number of chicks entering pullet house s in period t
A_{sgt}	The number of chickens transferred from pullet houses to hen house g in period t
INV_t	Egg inventory at the end of period t
X_t	The number of egg shortages in period t

Binary variables

Y_{st}	Takes value 1 if the chicks are assigned to chicken house s in period t
QA_t	Takes value 1 if chick ordering takes place in the poultry farm at time t
Z_{gt}	Takes value 1 if hens are assigned to hen house g in period t

By using the introduced symbols, the stated problem can be mathematically formulated as follows:

$$\begin{aligned}
MinZ = & \underbrace{\left(\sum_{t \in T} Cf_q \times QA_t \right)}_1 + \underbrace{\left(\sum_{s \in S} \sum_{t \in T} A^P \times Y_{st} + \sum_{g \in G} \sum_{t \in T} A^H \times Z_{gt} \right)}_2 + \\
& \underbrace{\left(\sum_{s \in S} \sum_{t \in T} K^P \times H_{st} + \sum_{s \in S} \sum_{g \in G} \sum_{t \in T} K^H \times A_{sgt} \right)}_3 + \underbrace{\left(\sum_{s \in S} \sum_{g \in G} \sum_{t \in T} Cd_{sg} \times A_{sgt} \right)}_4 + \\
& \underbrace{\left(\sum_{t \in T} (EI \times INV_t + KU \times X_t) \right)}_5
\end{aligned} \quad (1)$$

Subject to

$$QA_t \geq Y_{st}, \forall s, t \quad (2)$$

$$\sum_{g \in G} A_{sgt+K} = H_{st}, \forall s, t \quad (3)$$

$$INV_t = INV_{t_0} + \sum_{g \in G} L_{r=1} \times H_{st-K} - DEM_t + X_t, t = K + 1 \quad (4)$$

$$INV_t = INV_{t-1} + \sum_{s \in S} \sum_{r|t-K-r+1 > 0} L_r \times H_{st-K-r+1} - DEM_t + X_t, \forall t > K + 1 \quad (5)$$

$$Y_{st+l} \leq 1 - Y_{st}, \forall s, t, l = 1, 2, \dots, K + U^P \quad (6)$$

$$Z_{gt+l} \leq 1 - Z_{gt}, \forall g, t, l = 1, 2, \dots, R + U^H \quad (7)$$

$$\sum_{s \in S} H_{st} \leq YC, \forall t \quad (8)$$

$$H_{st} \leq C_s^P \times Y_{st}, \forall s, t \quad (9)$$

$$A_{sgt} \leq C_g^H \times Z_{gt}, \forall s, g, t \quad (10)$$

$$A_{sgt+K} \leq C_g^H \times Y_{st}, \forall s, g, t \quad (11)$$

The problem of egg production planning aims to minimize total cost over the development plan horizon (as shown in Eq. (1)). The first and second parts are related to the fixed cost of ordering chicks and the cost of using and cleaning pullet and hen houses, respectively. The third part is related to the cost of feeding chickens and chickens, the fourth part is related to the cost of transporting chickens and chickens to poultry farms and slaughterhouses and the fifth term is related to the cost of inventory and unmet demand.

Constraint (2) ensures that if none of the pullet houses are used in period t , no chicken order will be made. Constraint (3) guarantees that all chickens that enter hall g in period t can be moved to all accessible broiler houses s without any

losses after K weeks. Constraints (4) and (5) represent the balance between inventory level and demand for produced eggs in period t . In these constraints, chickens that enter the breeding process in period $t-K$ reach the laying age in period t and lay eggs at the rate of $L_1, L_2, \dots$, and L_R , respectively, until R weeks. Constraint (6) ensures that if pullet-house s is used in period t ($Y_{st} = 1$), it will be unavailable until $K + U^P$ weeks later, where U^P is the time needed to clean the hall. Similarly, constraint (7) ensures that if hen-house g is used in period t , it will be unavailable until $R + U^H$ weeks later, where U^H is the time needed to clean a hen-house. Based on the constraint (8), the total number of hatchings in period t in all pullet houses should not exceed the capacity of the slaughterhouse because all these chickens $K+R$ weeks later will be transferred to the slaughterhouse simultaneously. Condition (9) ensures that if hatching is not done in pullet house s in period t , the binary variables Y_{st} and H_{st} will be zero; otherwise, the number of hatchings should not exceed the capacity of the pullet house. Constraints (10) and (11) similarly apply identical conditions to the hen houses.

In the proposed model, indices $k=1, 2, \dots, K$ for the H_{st} and Y_{st} variables, as well as $r=1, 2, \dots, R$ for the A_{sgt} and z_{gt} variables, along with their respective constraints, have been eliminated, and thus, the number of variables and constraints has decreased in comparison to the previous analogous model.

3 Stochastic modeling of poultry industry problem and egg production planning

Egg production is a relatively long-term process, and there is a gap of several weeks between the time of hatching and the laying of eggs by the raised chickens, and it is pretty difficult to accurately determine the amount of demand for eggs at the time of hatching. Customers will be unsatisfied, and the company's reputation will suffer if the volume of produced eggs is below the level of actual consumer demand and maintenance costs will rise and eggs will rot if output exceeds market demand. Predicting customer behavior and using random planning methods can significantly reduce these costs. It is possible to establish a safe level for product production

by using potential restrictions if it is possible to predict the behavior of market demand in the form of a stochastic variable with a specific probability distribution function.

To deal with the uncertainty of the demand parameter, five methods with a new approach are presented below. Each of these methods provides a plan for egg production to the decision maker, and the number of eggs produced is obtained outside of the original model. The number of eggs obtained from each of these five methods is placed in the main model instead of customer demand parameter, and the deterministic model run and production planning will be determined.

Let $GE = (GE_1, GE_2, \dots, GE_T)$ represent the number of eggs produced in different weeks, and let $\tilde{D} = (\tilde{D}_1, \tilde{D}_2, \dots, \tilde{D}_T)$ represent the uncertain demand for eggs in those weeks. If the probability distribution function of the random variable is known, the various methods described below can be used to establish the weekly production level in such a way that the likelihood of a shortage or excess of inventory is minimally affected, as feasible. In these methods, the desired production amount is first calculated with different attitudes. Then in the optimization model presented in the previous section, these values are replaced by the demand parameter, and by solving the model, a production plan corresponding to that method is obtained.

Planning based on average data (Plan 1): One of the banal methods for dealing with random data is to use their average. In this approach, the average is employed to execute the deterministic model rather than considering all or some of the potential outcomes for the value of the random data. Birge and Louveaux (2011) with some actual examples showed that this method is not very efficient. In this method, egg demand is set equal to the average of the data. By substituting these numbers in constraints (4) and (5), a production plan will be provided for decision-makers.

Planning based on chance-constraint (Plan 2): In this method, the production plan is determined so that the probability of encountering a shortage is not lower than a certain confidence level. In other words, with a high probability, the amount of production should be more than the amount of demand. If $\tilde{D}_t$ and GE_t are uncertain demand and the amount of egg production in week t respectively, the possible condition will be as

$$pr\{GE_t \geq \tilde{D}_t\} \geq \alpha, \forall t \quad (12)$$

where $\alpha \in [0, 1]$ determines the confidence level of the constraint being met.

If $\tilde{D}_t$ has a certain cumulative distribution function $F_{\tilde{D}_t}$, the deterministic equivalent of the probability constraint can be written as follows:

$$pr\{\tilde{D}_t \leq GE_t\} \geq \alpha \Leftrightarrow F_{\tilde{D}_t}(GE_t) \geq \alpha \Leftrightarrow GE_t \geq F_{\tilde{D}_t}^{-1}(\alpha), \forall t \quad (13)$$

If weekly demand is considered to have a normal distribution with mean μ_t and standard deviation σ_t ($\tilde{D}_t \sim \mathcal{N}(\mu_t, \sigma_t^2)$), inequality (13) is written as follows:

$$GE_t \geq \mu_t + \sigma_t \varphi^{-1}(\alpha), \forall t \quad (14)$$

where φ is the cumulative distribution function of the standard normal variable.

This planning for egg production has more calculations compared to plan 1. In this program, because the customer's demand for eggs during the period (all weeks) is more than the average amount of demand in the entire period, therefore the model faces a very low shortage of eggs and a very high surplus of eggs. Therefore, in the next proposed programs, the goal is to create more balance and proportionality between shortage and surplus.

Planning based on the aggregation of demands (Plan 3): The attitude used in the previous method to handle uncertainty was conservative, assuming that demand would be higher than average in all weeks and that egg stock should always be kept at a high level to meet customer demand. This approach necessitated ordering more chickens, which increased costs related to ordering, feeding, maintaining poultry houses and chicken coops, and transportation. In practice, demand may be lower than the mean value during certain periods, leading to surplus inventory that can be used to meet demand in subsequent weeks. To account for this potential surplus, one way is to treat the total demand as a random variable and adjust accordingly in inventory planning for future weeks. If the weekly demands are independent of each other and $\tilde{D}_t \sim \mathcal{N}(\mu_t, \sigma_t^2)$, then the sum of the demands of the first n weeks will be a normal variable in the form of $\sum_{t=1}^n \tilde{D}_t \sim \mathcal{N}(\sum_{t=1}^n \mu_t, \sum_{t=1}^n \sigma_t^2)$. In this case, the amount of weekly production is determined step by step as follows:

Step 1: Using Eq. (14), we get the number of eggs produced in the first week ($t=1$) as follows:

$$\tilde{D}_1 \sim \mathcal{N}(\mu_1, \sigma_1^2) \rightarrow GE_1 = \mu_1 + \sigma_1 \varphi^{-1}(\alpha),$$

Step 2: Using Eq. (14) and step 1, we get the number of eggs produced in the second week ($t=2$) as follows:

$$\begin{aligned} \tilde{D}_1 + \tilde{D}_2 &\sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \\ \rightarrow GE_1 + GE_2 &\geq (\mu_1 + \mu_2) + (\sqrt{\sigma_1^2 + \sigma_2^2}) \varphi^{-1}(\alpha) \\ \xrightarrow{GE_1 = \mu_1 + \sigma_1 \varphi^{-1}(\alpha)} GE_2 &= \mu_2 + (\sqrt{\sigma_1^2 + \sigma_2^2} - \sigma_1) \sigma^{-1}(\alpha) \end{aligned} \quad (15)$$

Step n: Using Eq. (14) and step n-1, we get the number of eggs produced in the n^{th} week as follows:

$$GE_n = \mu_n + (\sqrt{\sum_{t=1}^n \sigma_t^2} - \sqrt{\sum_{t=1}^{n-1} \sigma_t^2}) \varphi^{-1}(\alpha) \quad (16)$$

If for all t , $\mu_t = \mu$ and $\sigma_t^2 = \sigma^2$ then:

$$GE_n = \mu + (\sqrt{n} - \sqrt{n-1}) \sigma \varphi^{-1}(\alpha) \quad (17)$$

Planning based on considering the average surplus cyclically (Plan 4): In this approach, after determining the production amount of the first week, the expected excess inventory corresponding to this production level can be calculated, and this amount can be included in the inventory of the following weeks.

The expected surplus value is derived from the following relations if the weekly demand is a random variable with the probability density function $f_{\tilde{D}}(t)$ and the amount of production is equal to G :

$$\hat{L}(G, \tilde{D}) = \int_{-\infty}^{+\infty} \max(G-t, 0) f_{\tilde{D}}(t) dt = \int_{-\infty}^G (G-t) f_{\tilde{D}}(t) dt \quad (18)$$

Lemma 1 If $\tilde{D} \sim \mathcal{N}(\mu, \sigma^2)$, the expected surplus is obtained as follows:

$$\hat{L}(G, \tilde{D}) = \sigma \left(\varphi\left(\frac{G-\mu}{\sigma}\right) + \phi\left(\frac{G-\mu}{\sigma}\right) \frac{G-\mu}{\sigma} \right) \quad (19)$$

where $\varphi(\cdot)$ and $\phi(\cdot)$ are the probability density function and cumulative distribution function of the standard normal distribution, respectively.

This planning is performed cyclically and based on the surplus consumption of the preceding cycle in the subsequent cycle in two stages. In the first stage, using Eq. (14), the definite demand ($\tilde{D} = GE$) is calculated with a confidence level of $\alpha\%$ for the initial cycle. Each cycle consists of R weeks of egg laying, so in the first stage, Eq. (18) calculates the surplus for each R week. In the second stage, each surplus obtained in the R week of the first cycle is consumed in the R week of the second cycle, respectively. Accordingly, the definite demand for the R weeks of the second cycle is obtained by subtracting the surplus obtained from the R weeks of the first cycle from the value obtained using Eq. (14). This two-step process is applied to the entire planning horizon. For instance, in the third cycle, the demand for each R -week period is computed using chance constraints with a confidence level of $\alpha\%$, and the surpluses are used in the R -week period of the fourth cycle. This process determines the definite demand over the entire time horizon.

Step 1: In the first cycle, using Eq. (14), we get the number of eggs produced as follows: $\tilde{D}_t \sim \mathcal{N}(\mu, \sigma^2) \rightarrow GE = GE_t = \mu + \sigma \varphi^{-1}(\alpha)$, $t = 1, 2, \dots, R$

Step 2: In the second cycle, using Eq. (18), the expected surplus value of the first cycle (step 1) is calculated, which will be consumed in the second cycle. We get the number of eggs produced as follows: $GE_t = \mu + \sigma \varphi^{-1}(\alpha) - \hat{L}(GE, \tilde{D})$, $t = R+1, R+2, \dots, 2R$

Step 3: In the third cycle, using Eq. (14), we get the number of eggs produced as follows: $\tilde{D}_t \sim \mathcal{N}(\mu, \sigma^2) \rightarrow GE = GE_t = \mu + \sigma \varphi^{-1}(\alpha)$, $t = 2R+1, 2R+2, \dots, 3R$

Step 4: In the fourth cycle, using Eq. (18), the expected surplus value of the third cycle (step 3) is calculated, which will be consumed in the fourth cycle. We get the number of eggs produced as follows: $GE_t = \mu + \sigma \varphi^{-1}(\alpha) - \hat{L}(GE, \tilde{D})$, $t = 3R+1, 3R+2, \dots, 4R$

In the same way, the planning of egg production is done cyclically and based on the surplus consumption of the previous cycle in the next cycle in two stages. In the first stage, the number of eggs produced is determined based on the relationship (14), and in the second stage, the expected surplus of cycle eggs is calculated and consumed using (18). That is, the number of eggs produced is obtained by subtracting the surplus of the first stage from the amount of eggs produced in the second stage.

Planning based on considering the average surplus consecutively (Plan 5): This method is similar to the previous approach, with the exception that in the previous method, the surplus for even cycles (e.g., the second and fourth week of each cycle) was not calculated. In contrast, this method calculates the average surplus for each R week based on the production level and subtracts it from the desired production amount for the subsequent cycle.

Table 1 shows a summary of the introduction and basic differences of the five plans for egg production.

4 Computational experiments

In this section, the model is initially implemented under deterministic data conditions, and the resulting output from applying the model to multiple farms of varying sizes is compared to that of Boonmee and Sethanan (2016). Next, we utilize the methods described in the previous section to address demand uncertainty, and present five different production plans. The data used in this research is based on the data taken from the article by Boonmee and Sethanan (2016). This data was obtained from a semi-industrial manufacturing company in north-eastern Thailand.

Table 1 Summary of five egg production plans

	Important features	Weekly production level	Operational performance in terms of shortage and excess
Plan1	Weekly demand is equal to long-term average demand	The production level is equal to the average demand	Shortage and surplus evenly in different weeks
Plan2	Determining the production level so that the demand can be met with high reliability	The production level is much higher than the average demand	The lowest total shortage and at the same time the highest total excess
Plan3	Considering the total accumulated demand from the beginning to the current week as a random variable	The production level is decreasing during the time horizon and its value is always between the production level in plan 1 and 2	It has the best performance in terms of the cost ratio of shortage to surplus with different coefficients
Plan4	Compared to plan 2, the average surplus of the previous weeks is periodically reduced from the production level	Compared to the plan 2, the production level decreases once every three weeks and returns to the initial amount	Compared to plan 1, there is less of an egg shortage, and compared to plan 2, there is less of an egg surplus
Plan5	Compared to plan 2, the average surplus of the previous weeks is consecutively reduced from the production level	Compared to plan 2, the production level decreases and increases again every three weeks	Compared to plan 1, there is less of an egg shortage, and compared to plan 2, there is less of an egg surplus

To implement the proposed model, a farm with $|S|$ pullet houses and $|G|$ hen houses is considered. Over $|T|$ weeks, various problems of different sizes are implemented, and the model runs for each instance. The required data for executing each problem number is defined using the pattern provided by Boonmee and Sethanan (2016) as follows:

$$DEM_t \in [50000, 275000], C_s^p \in [7000, 14000], C_s^h \in [4000, 15000]$$

$$K = 2, R = 3, L_r = 7, KU = 5, EI = 1$$

All models are implemented using the Cplex12 solver in the Aimms software environment and on Intel(R) Core(TM) i5-4200U CPU @ 1.6 GHz processor, with 3 GBytes of RAM, and within a maximum time of 1 h.

4.1 Implementation of the proposed model with deterministic data

The results obtained from the implementing of the proposed deterministic model and the previous model in different conditions are shown in Table 2. In the first three columns, the farm specifications, including the number of halls and slaughterhouse capacity, are given. In the following columns, the performance of the proposed model and previous model (reported in parentheses) is shown.

As can be seen in Table 2, the CPU time of the new model presented in this research in all modes 1 to 8 is much less than the original model in the previous article Boonmee and Sethanan (2016). For instance, the previous

model took 19,100 s to arrive at the optimal solution for mode 8 while the proposed model only needed 1.01 s to reach the same conclusion. In the previous model, instances 9 to 12 could not be solved. However, in the modeling presented in this research, problem number 9 was solved in 625 s, and the exact answer was obtained. Also, problem numbers 11 - 12 have been solved with a meager gap in one hour. However, in the previous model, a suitable solution for the optimization problem could not be found within one hour, likely due.

to the large size of the model. This indicates that the presented model is much better for the given data because it usually obtains the exact solution with a much lower CPU time.

4.2 Comparison of proposed methods for dealing with demand uncertainty

Suppose a farm with the specifications of problem number 9, introduced in Table 2, is available, and the weekly demand for eggs follows a non-deterministic normal distribution given by $\tilde{D}_t \sim N(\mu = 162500, \sigma^2 = 20000^2)$. A 90% confidence level is considered to avoid egg stock shortages ($\alpha = 0.9$). To address this uncertainty, the five

Table 2 Evaluating the performance of the proposed model and the previous model in situations with certain demands

Problem number	Specifications of the problems		Performance of the proposed model and previous model (reported in parentheses)			
	(S , G , T)	YC	# Binary variables	#Constraints	CPU time (sec)	Gap (%)
1	(5,8,5)	10,000	55 (240)	562 (1888)	0.20 (3)	0 (0)
2	(5,8,5)	20,000	55 (240)	562 (1888)	0.16 (4)	0 (0)
3	(8,15,5)	10,000	99 (425)	1387 (4758)	0.09 (7)	0 (0)
4	(8,15,5)	20,000	99 (425)	1387 (4758)	0.17 (6)	0 (0)
5	(5,8,10)	10,000	120 (480)	1257 (3763)	0.48 (55)	0 (0)
6	(5,8,10)	20,000	120 (480)	1257 (3763)	0.28 (22)	0 (0)
7	(8,15,10)	10,000	214 (850)	3097 (10,333)	1.64 (9715)	0 (0)
8	(8,15,10)	20,000	214 (850)	3097 (10,333)	1.01(19,100)	0 (0)
9	(15,25,20)	30,000	770 (2920)	17,637 (52,860)	625 (b)	0 (-)
10	(15,25,30)	30,000	1170 (4380)	26,907 (79,270)	3600 (b)	0.13 (-)
11	(20,40,20)	40,000	1160 (4420)	35,437 (106,660)	3600 (b)	0.23 (-)
12	(20,40,30)	40,000	1760 (6630)	54,057 (184,790)	3600 (b)	0.34 (-)

b = Problem is too large to be solved by MPL/CPLEX

methods presented in Section 4 are employed to calculate the desired production amount, which is then incorporated as the demand parameter in the deterministic model. The output of the optimization model for each of these methods provides decision-makers with a plan for chicken breeding, egg production, and other relevant factors. Figure 2 shows the number of eggs produced per week according to the different plans (assuming $K=2$, so egg production commences from the third week). As shown in this figure, in most weeks, the lowest production is related to plan 1, and the highest production is linked to plan 2, and production levels in the other three plans falling between these two extremes.

This section compares and evaluates the egg production plans developed using the discussed methods. The comparison is performed by calculating the amount of shortage and surplus obtained from each plan. One hundred random data samples for the demand in each week are generated

normally, with an average of 162,500 and a standard deviation of 20,000. The random data used in this research were generated based on the range of real data observed in a semi-industrial manufacturing company in Northeast Thailand. By considering the warehouse input (obtained from the deterministic model) and the warehouse output (randomly generated data), the amount of shortage or surplus can be calculated every week. The excess cost is always set to 1, and the shortage cost equals to KU. Table 3 presents the average total number of shortages and surpluses for each plan. Plan 2 has the slightest shortage and the most considerable surplus, while Plan 1 has the most considerable shortage and the slightest surplus. The third and fourth rows of the table report the ratios of the total shortage and surplus to the total demand for the one hundred data samples. These data indicate that Plan 2 has the slightest shortage and the most considerable surplus.

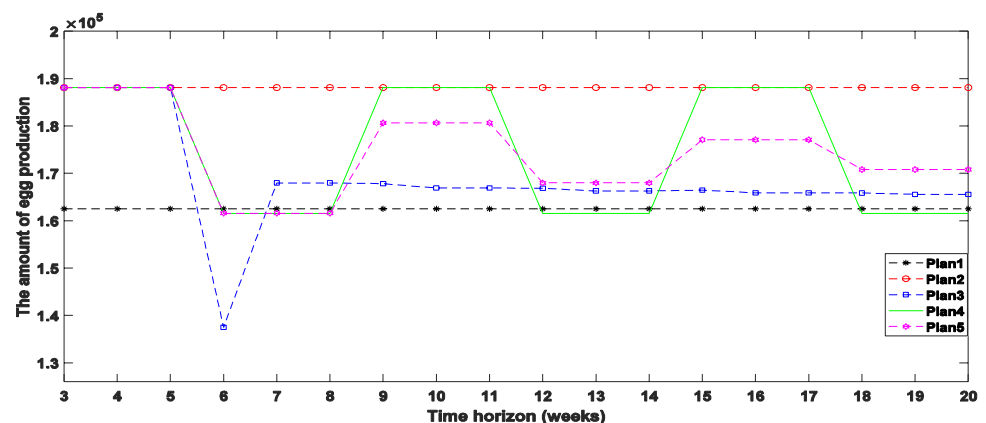
Fig. 2 Weekly egg production according to different plans for $\alpha=0.90$ 

Table 3 The average of shortage and surplus for 100 data samples for different plans for $\alpha=0.90$

	Plan 1	Plan 2	Plan 3	Plan 4	Plan 5
The average of shortages	56,686.95	1872.83	11,063.71	3318.70	3472.60
The average of surpluses	689,463.91	4,405,608.36	1,525,987.30	2,514,400.00	2,314,300
The average ratio of total shortage to total demand	0.0191	0.00063452	0.0037	0.0011	0.0012
The average ratio of total surplus to total demand	0.2404	1.5716	0.5297	0.8692	0.8006

Table 4 The average cost of shortage and surplus for different plans for $\alpha=0.90$

KU	Plan1	Plan2	Plan3	Plan4	Plan5
1	1,754,150	5,569,100	2,597,751	3,599,750	3,396,945
5	1,980,900	5,576,600	2,641,951	3,612,950	3,393,445
20	2,831,200	5,604,700	2,807,951	3,662,750	3,462,845
50	4,531,800	5,660,800	3,139,851	3,762,350	3,567,045
100	7,366,200	5,754,500	3,693,051	3,928,250	3,740,645
1000	58,384,000	7,440,000	13,650,651	6,915,150	6,865,945

Table 4 displays the average total cost of shortage and surplus for various KU values, along with the fixed cost of each plan. The fixed costs of the objective function are 1,008,000, 1,161,600, 1,060,651, 1,081,950, and 1,079,145 for plans 1–5, respectively. According to Table 5, plan 1 for KU = 1 and 5 has the lowest objective function cost, while plan 3 has the lowest objective function cost for KU = 20, 50, and 100. Plan 5 has the lowest objective function cost for KU = 1000.

Based on the above tables, Plan 2 is the best option for egg production under uncertainty if the decision-makers goal is to meet customer demand without any shortages. Plans 3, 1, and 5 are more appropriate for producing eggs with uncertain demand if the objective is to meet demand with the least amount of shortage and surplus, and hence at the lowest cost, for all KU values. To provide a more precise comparison of these methods in terms of shortage, surplus, and the average total cost of eggs, some charts are presented for plans 1, 3, and 5, based on 100 randomly generated data samples, following a normal distribution within the specified range.

Figure 3 shows that the ratio of the total egg shortage to the total demand of one hundred data samples. According to this figure if one of these three production programs is implemented in the farm, the expectation of facing a shortage of inventory is the highest and lowest in plans 1 and 5, respectively. If the decision maker's goal is to meet only the customer's demand with the least shortage, plan 5 is the best plan. However, as shown below, with the implementation of this plan, there is a high possibility of facing excess inventory and its corruption.

Figure 4 displays the surplus over the last week for 100 data samples. This figure shows that the surplus of eggs is more in plans 5, 3, and 1, respectively. Figure 5 shows the ratio of the total surplus of eggs to the total demand of the data sample. In this figure, it is evident that the identical issue is clearly depicted, with plans 1, 3, and 5 showing the slightest surplus, respectively.

According to this figure, plan 1 has the lowest surplus of eggs compared to plans 3 and 5, and plan 5 has the highest surplus of eggs compared to plans 1 and 3 on weekends. Therefore, plan 3 can be more appropriate and acceptable in terms of excess cost and customer demand compared to plans 1 and 5. If the goal of the decision maker is to meet the customer's demand with the least amount of storage (surplus eggs), plan 1 is the best because, in this plan, egg production is less than plans 5 and 3. In Plan 5 because the production of eggs is high in all weeks, therefore it faces a much larger surplus of eggs compared to plans 1 and 3. Also, plan 3 has more surplus compared to plan 1, and has a smaller surplus compared to Program 5. Therefore, in general, if we take into account the costs of shortage and surplus and meeting customer demand, it is concluded that program 3 is more suitable compared to plans 1 and 5. The average total cost of

Table 5 The average of shortage and surplus for 100 data samples for different plans for $\alpha=0.95$

	Plan 1	Plan 2	Plan 3	Plan 4	Plan 5
The average of shortages	56,686.95	819.70	4798.97	1248.50	1248.50
The average of surpluses	689,463.91	5,634,913.38	1,853,794.94	3,242,100	3,045,300
The average ratio of total shortage to total demand	0.0191	0.00027736	0.0016	0.00041880	0.00041880
The average ratio of total surplus to total demand	0.2404	1.9391	0.6426	1.1188	1.0513

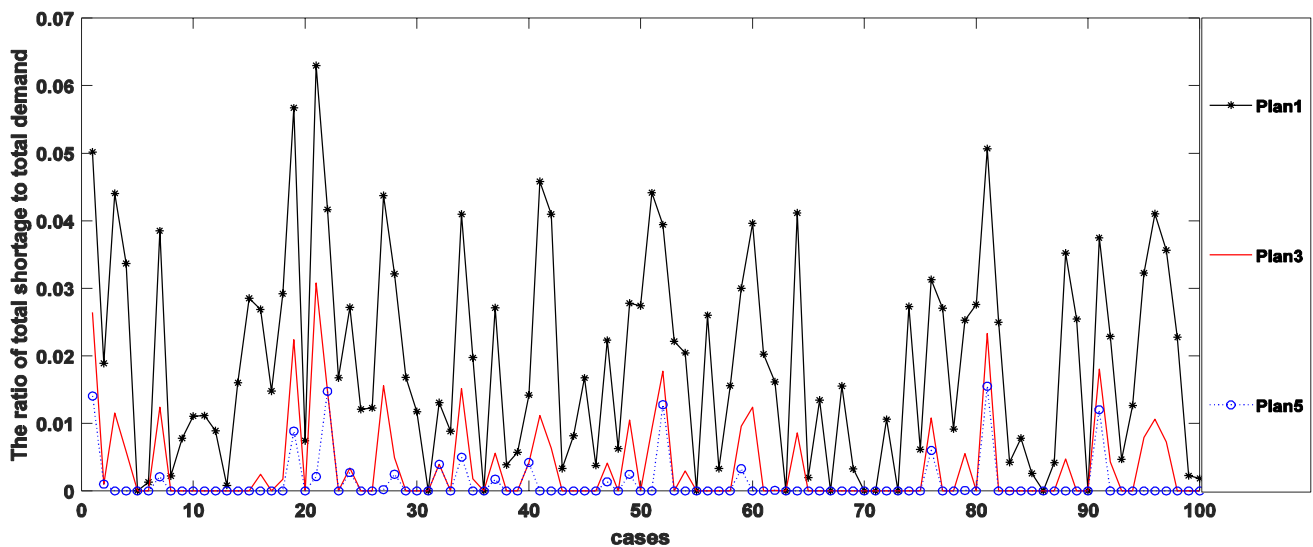


Fig. 3 The ratio of the total shortage to total demand of the given sample is $\alpha=0.90$

the egg surplus and shortage for $KU=50$ is shown in Fig. 6. This figure shows that the average total cost of egg shortage and surplus in plans 3 and 5 is much lower than in plan 1. In the real world, in a supply chain, the goal is to fulfill customer demand with the least shortage and the lowest cost, so the cost of shortage should be considered more than the cost of storage. Therefore, according to the observations of the tables and figures, plans 3 and 5 are more suitable for egg production to face uncertainty in terms of costs of shortage and surplus of eggs.

In the following, five methods are compared for $\alpha=0.95$. In plans 2, 3, 4, and 5, the number of eggs increases with the increase in confidence level (α) because, according to Eqs. (12)–(14), the number of eggs produced increases with the increase of confidence level. Therefore, the quantity of shortage and surplus in the five provided plans may change, as could the sum of the average expenses of the entire shortage and surplus. These changes are shown in Tables 5 and 6 and Figs. 8, 9, and 10. As a result, with the increase in the level of confidence, only the amount of shortage and surplus

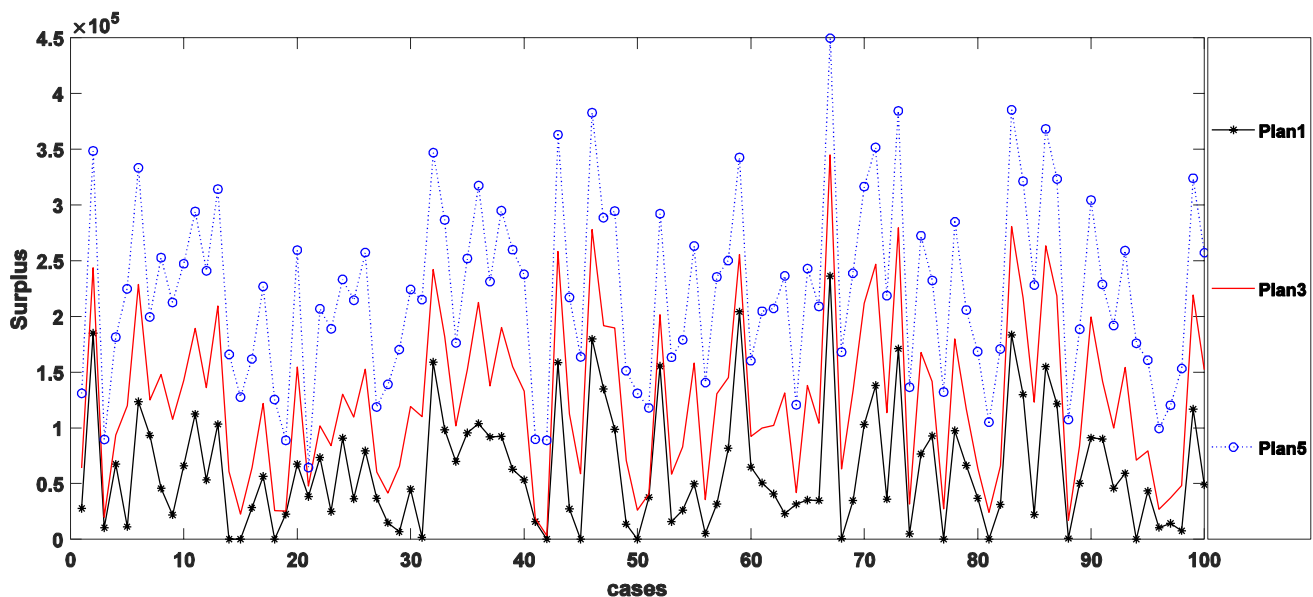


Fig. 4 The surplus in the last week for $\alpha=0.90$

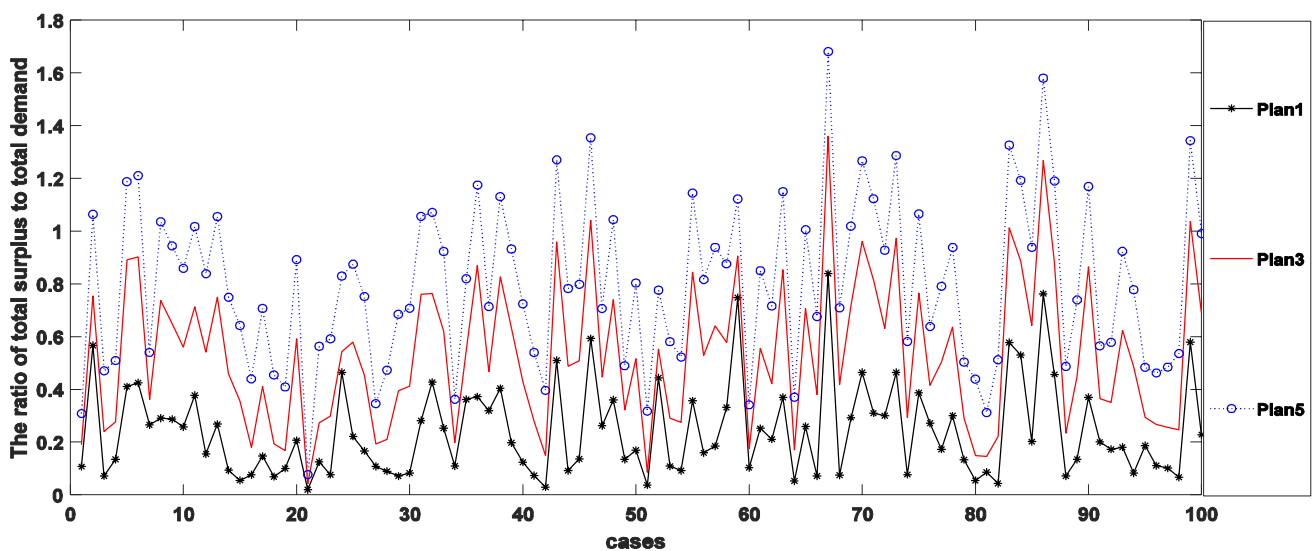


Fig. 5 The ratio of the total surplus to total demand of the given sample for $\alpha = 0.90$

and the total costs change. However, in general, no noticeable change has been seen, and plans 3, 1, and 5, respectively, in terms of the amount of shortage and surplus of eggs, as well as the total amount of costs for egg production planning is more suitable than other programs.

Figure 7 shows the planning of weekly production for the five presented plans. In general, it can be seen that from confidence level 0.90 to 0.95, the amount of production has increased in plans 2 - 5.

Table 5 demonstrates that for plans 2 - 5, as the confidence level rises from 0.90 to 0.95, there will be less of a shortage of eggs and more of a surplus.

For plans 1 through 5, the fixed costs of the objective function are 1,008,000, 1,136,258, 1,073,664, 1,105,437, and 1,103,771, respectively. Table 6 shows that in plans 2 - 5, with the increase of the confidence level from 0.90 to 0.95, the average of objective function values increases.

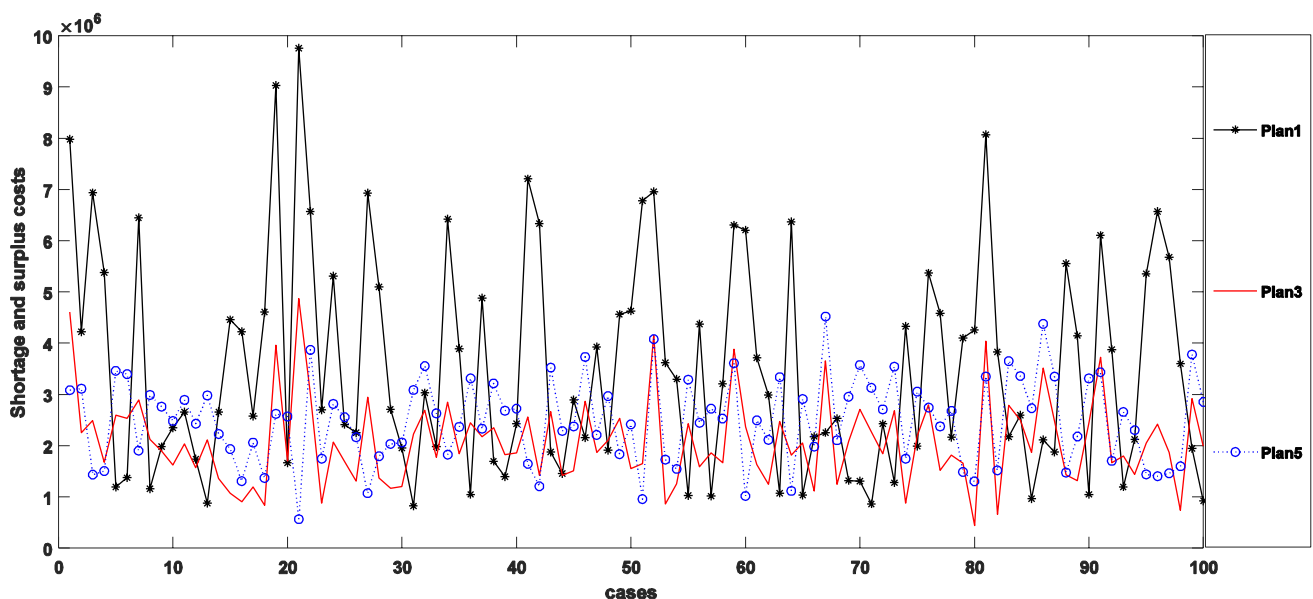


Fig. 6 The average cost of shortage and surplus for $KU = 50$ and $\alpha = 0.90$

Table 6 The average cost of shortage and surplus for different plans for $\alpha=0.95$

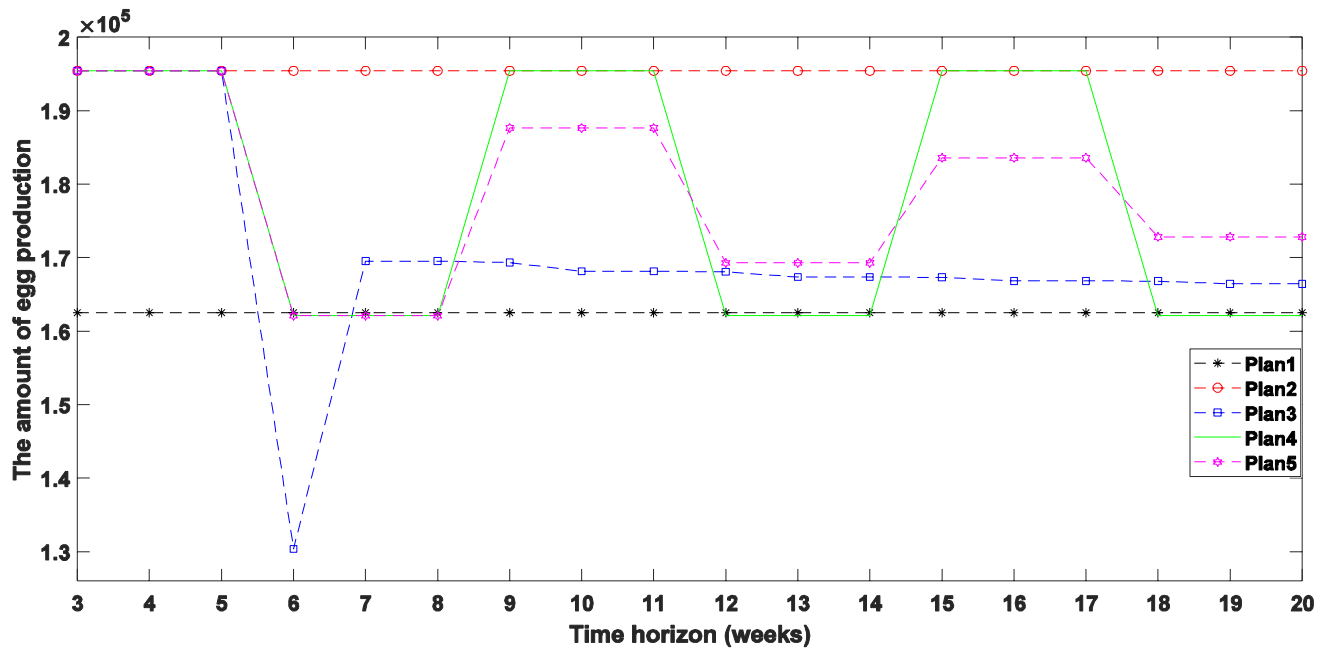
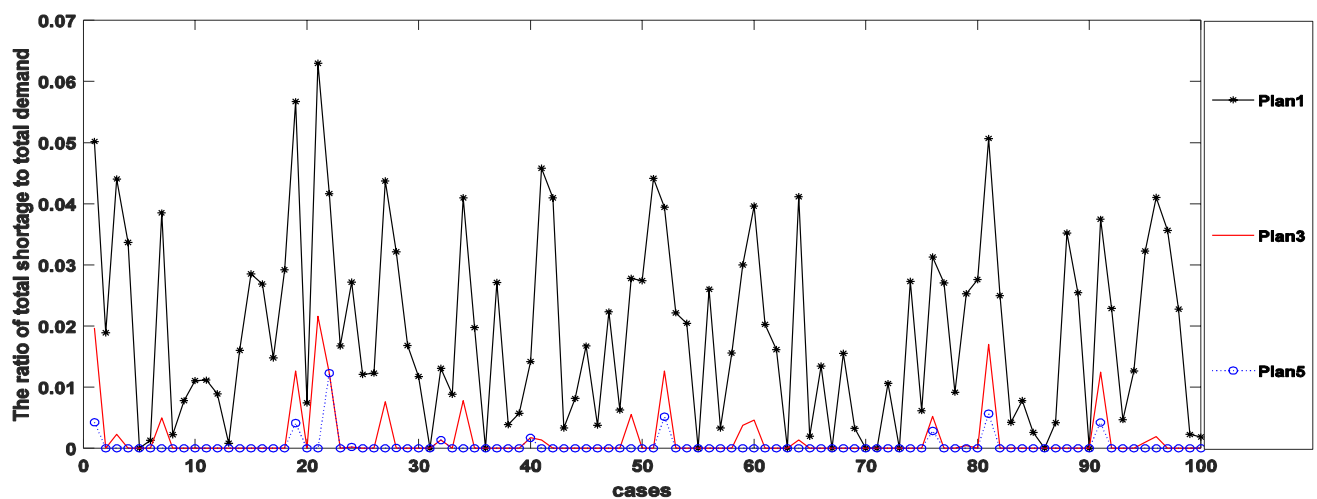
KU	Plan1	Plan2	Plan3	Plan4	Plan5
1	1,754,150	6,841,082	2,932,264	4,348,737	4,150,271
5	1,980,900	6,844,382	2,951,464	4,353,737	4,155,271
20	2,831,200	6,856,682	3,023,464	4,372,437	4,174,071
50	4,531,800	6,881,282	3,167,364	4,309,937	4,211,471
100	7,366,200	6,922,282	3,407,364	4,472,337	4,273,871
1000	58,384,000	7,659,982	7,726,464	5,595,937	5,397,471

According to Fig. 8, the most shortage is observed in plan 1, and plans 5 and 3 have the most minor shortage, respectively.

According to Figs. 9 and 10, plans 1, 3, and 5 have the lowest surplus of eggs, respectively.

According to Fig. 11, the average total cost of egg shortage and surplus in Plans 3 and 5 is much lower than in Plan 1.

Finally, it can be concluded that for both levels of confidence considered, if decision-makers assign a high importance coefficient to avoiding shortages, production plan 5 can meet this

**Fig. 7** production according to different plans Weekly egg for $\alpha=0.95$ **Fig. 8** The ratio of the total shortage to total demand for $\alpha=0.95$

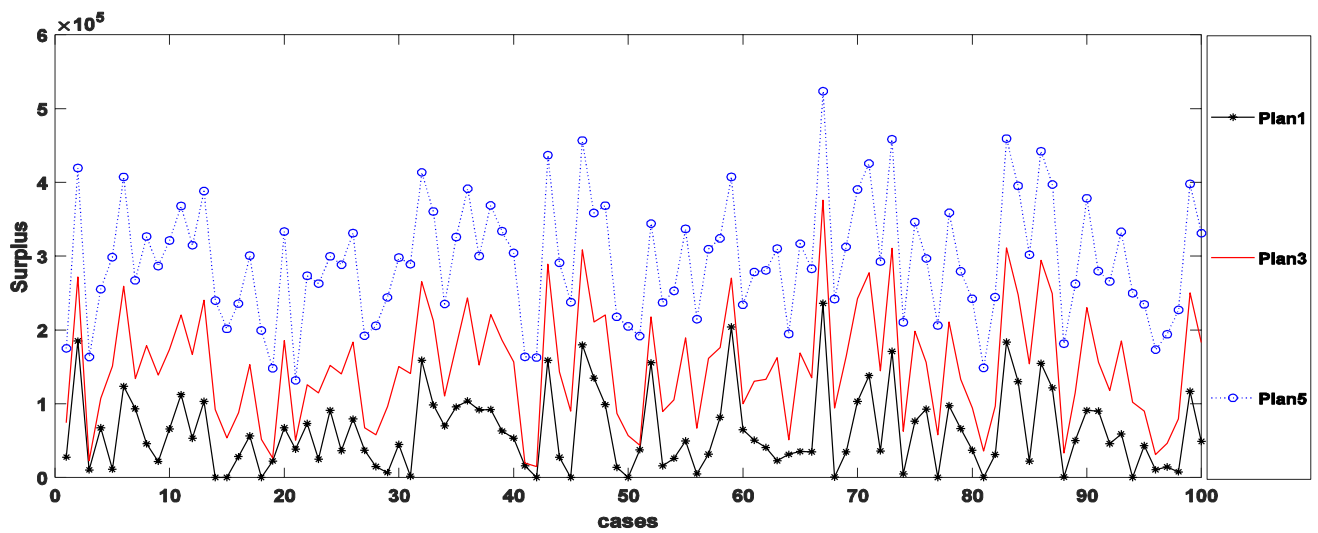


Fig. 9 The surplus in the last week for $\alpha=0.95$

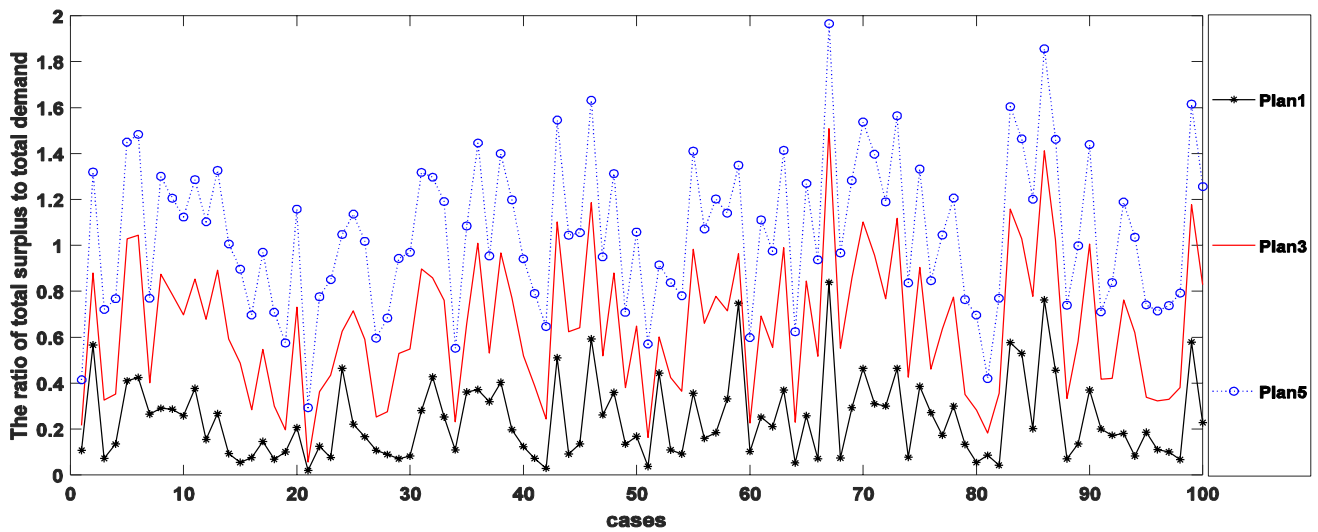


Fig. 10 The ratio of the total surplus to total demand for $\alpha=0.95$

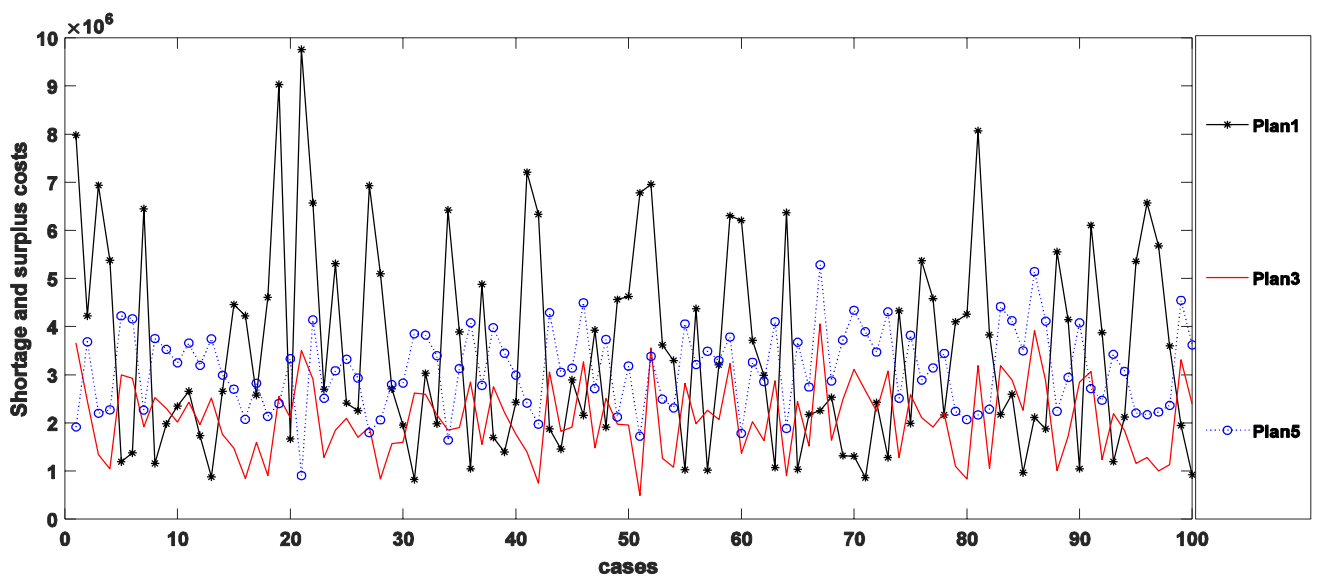


Fig. 11 The average of total shortage and surplus cost for $KU=50$ and for $\alpha=0.95$

criterion. However, when taking into account both the costs of shortages and surpluses simultaneously, plan 3 demonstrates a more balanced performance compared to the other plans.

5 Conclusions

In this article, an investigation was carried out on a laying hen breeding farm, which includes several pullets and hen houses. The objective is to determine the optimal allocation of hatching in the pullets in different weeks, in order to ensure sufficient egg production at the laying age while minimizing the cost of production, transportation, storage, and avoiding supply-demand imbalances. Previously, this problem was formulated under the assumption of certain demand, leading to a complex model and an imprecise solution method. However, in this study, we have refined the problem formulation to reduce constraints and binary variables, thus enabling the model to find an optimal solution quickly. In this problem the interval from hatching to laying and slaughter of old hens are represented as K and R weeks, respectively, and thus in new MILP model, the formulation was simplified by omitting the variable and constraints related to age. To evaluate the performance of the proposed model, we compared it with a similar previous model using real data introduced by Boonmee and Sethanan (2016). The results demonstrated the efficiency of the new model in terms of CPU time and the optimality of the solutions. It is found that the proposed model can be executed with significantly less CPU time, providing an exact solution for large datasets such as case 9 in Table 2, and a negligible time gap within an hour for cases 10, 11, and 12.

Furthermore, we have also incorporated considerations for demand uncertainty by utilizing stochastic planning and chance constraints method to develop various production plans for weekly egg production. The aim is to mitigate shortages and surpluses as much as possible while meeting customer demand. This study considers the uncertainty in demand for produced eggs and proposes a new approach, using chance constraints and expected value of surplus to deal with this uncertainty. Five methods are presented for dealing with stochastic optimization based on mean demand and chance constraints. Some of these plans are based on surplus egg consumption in previous periods. These methods are used independently of the main model and without affecting the CPU time of the main problem to deal with demand uncertainty, providing egg production plans for decision-makers. To evaluate and compare these different production plans one hundred randomly generated normal data samples were produced, and the total shortage, surplus eggs, and average total shortage and surplus costs were calculated based on different cost coefficients for the shortage. According to the obtained results, planning methods 3 and 5 are more suitable

for meeting customer demand with the minimum shortage, surplus, and overall cost of shortage and surplus.

For future works, it is suggested to study farms with larger dimensions including laying and broiler chickens, and also the uncertainty of other parameters such as the mortality rate of chickens can be taken into account in the modeling of the problem. Also, providing other planning methods such as machine learning and simulation can be beneficial.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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